

Lakshay Chauhan

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EDUCATION

Indraprastha Institute of Information Technology (IIIT) Delhi

Jul 2021 – Jun 2025

Bachelor of Technology in Computer Science and Engineering

Delhi, India

WORK EXPERIENCE

Member of Technical Staff

Jul 2025 – Present

ZL Technologies, Inc | [Website](#)

Hyderabad, India

- Worked on task synchronization and bulk parallel processing across clusters within ZL's analytics module, resulting in over a 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection when processing millions of items during data loading.

Research Assistant | [Gitlab Project Organization](#)

May 2024 – May 2025

Networks and Systems Security Lab - IIITD | [Website](#)

Delhi, India

- **Objective:** Address quality degradation in video streaming applications when fallback from QUIC to TCP occurs due to UDP blocking.
- **Solution:** Implemented a mechanism to send notifications in QUIC packets to anticipate UDP blocking. Server logs the kernel congestion window state to be used in the upcoming TCP connection. Improved user experience by avoiding TCP slow start, thus preventing buffering during fallback.

PROJECTS

Minimal Compiler Infrastructure | [Repository](#)

May 2024 – Present

- **Key Skills:** *Compiler Infrastructure, Design and Optimization, Rust, FASM, WASM, Graphviz*
- Developed a minimal compiler infrastructure inspired by LLVM. It converts source code into an optimized control flow graph (CFG) and translates it into target-specific assembly code.
- Implemented CFG optimization passes, such as identifier validation, constant folding, and CFG simplification, ensuring high-performance code generation.

Exploring Compiler Optimizations | [Repository](#)

Nov 2024 – Feb 2025

- **Key Skills:** *Compiler Optimization, Compiler Passes, Garbage Collectors, Type Safety, LLVM, C, C++*
- Developed LLVM-based compiler and runtime systems including a null pointer dereference detection pass, a conservative bump-pointer garbage collector for C with root scanning, and memory safety mechanisms enforcing spatial and weak type safety through bounds tracking and invalid pointer prevention.

Garbage Collector for C | [Repository](#)

Feb 2025 – Present

- **Key Skills:** *Runtime Systems, C, Garbage Collection, Memory Management, Systems Programming*
- Developed a conservative mark-and-sweep garbage collector for C with an 8-byte aligned bump allocator and segmented heap design.
- Implemented stack, heap, and global-root scanning for automatic memory reclamation.
- Designed page-level memory management using `mmap`, `mprotect`, and `madvise`.

TECHNICAL SKILLS

Languages: C, C++, Rust, Assembly, Java, Kotlin, JavaScript, Python, Bash, Haskell

Frameworks: PyTorch, Django, ReactJS, Numpy, Pandas, Tauri, LibGDX, Raylib

Developer Tools: Emacs, Linux, Git, GDB, Markdown, Google Cloud Platform, OpenLiteSpeed, SQLite3

Technical Electives: Data Structures & Algorithms, Operating Systems, Cryptography, Database Management, Computer Security, Computer Networks, Compilers, Machine Learning, Natural Language Processing

AWARDS

Awarded **Summer Undergraduate Research Fellowship** in 2023 by IRD-IIITD for the project “*Utilizing ultrasonic distance sensors as a mapping tool to design user-friendly CST*”.

Awarded **CHANAKYA Fellowship** by iHub Anubhuti Foundation in 2024 for the project “*A Unified Approach to User Emotion Detection through Emojis and Textual Analysis*”.